Article

CLOP-DiT: Text-Guided Single-Cell Gene Expression Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

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Abstract

Generating realistic synthetic single-cell gene expression profiles conditioned on structured text descriptions of cell identity would enable data augmentation and rare cell-type simulation, and could in future support hypothesis-driven perturbation studies. We present CLOP-DiT, a two-stage pipeline combining Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained via flow matching. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings into a shared 512-dimensional space (3.02M parameters), amplifying inter-type separation from 0.6% to 78% in cosine distance. DiT1D (22.10M parameters, 8 AdaLN-Zero blocks) then performs conditional flow matching in scGPT latent space, and a frozen scGPT decoder maps generated latents to gene expression. Across 100 cell types from 80 datasets (220,304 cells, 1,088 text groups), CLOP-DiT achieves *k*-nearest-neighbor accuracy of 36.9% (37× random chance, but 52 percentage points below the 89% real-data baseline) and steering accuracy of 81.0% (random = 50%) in the high-fidelity regime (CFG = 2.0, Euler solver); the recommended high-diversity regime (CFG = 1.0, Midpoint solver) reaches a diversity ratio of 0.93 at 80.7% steering. Unconditional generation collapses to random baselines, confirming that text conditioning is the sole driver of type specificity. In-distribution downstream biological validation—clustering alignment, classifier transfer (expression-space logistic regression accuracy 30.8%, macro F1 0.247), and differential expression concordance (mean logFC Pearson $r \approx 0.39$ across three contrasts, range 0.17–0.56)—demonstrates that generated cells partially integrate into standard single-cell workflows, though a discriminator AUC of 0.656 and the 52-point KNN gap from real data indicate substantial room for improvement. Expression-level validation across five tissues shows high decoder-reconstructed gene-mean fidelity (Pearson $r > 0.999$); however, per-gene variance correlation is near zero (r_{var} range: -0.019 to $+0.013$), indicating that cell-to-cell heterogeneity is not preserved—a critical limitation for applications requiring within-population variability.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; text-guided generation; flow matching; gene expression; foundation model

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1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of cellular heterogeneity at unprecedented resolution [1,2]. However, generating realistic synthetic single-cell expression profiles conditioned on structured text descriptions of cell identity remains an open challenge. Such a capability would enable data augmentation for rare

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cell types, and could potentially support hypothesis testing via *in silico* perturbation and atlas completion without additional sequencing, though these applications remain to be validated.

The clinical motivation for text-conditioned generation is threefold. First, rare cell populations—circulating tumor cells, therapy-resistant subclones, and transitional progenitor states—are chronically undersampled in standard scRNA-seq experiments, yet their accurate representation is critical for biomarker discovery and treatment-response prediction. Synthetic augmentation of these populations could improve classifier balance and statistical power without additional sequencing cost. Second, large-scale atlas efforts such as the Human Cell Atlas [3] require imputation of missing cell states across tissue contexts; a generative model conditioned on descriptive text could fill gaps where donor material is unavailable. Third, *in silico* hypothesis generation—predicting the transcriptomic consequences of a cell-state transition described in structured text—could accelerate experimental design in perturbation biology, although this application remains speculative and unvalidated.

A fundamental technical gap separates current generative methods from this vision. Variational autoencoders [4] such as scVI [5,6] and scGen [7], geometric deep generative models [8], and foundation models such as Geneformer [9] and scBERT [10–13] all condition on categorical cell-type labels or perturbation metadata. To our knowledge, no existing method accepts structured text biological descriptions—incorporating cell type, tissue, organism, marker genes, and disease context in a fixed template—as conditioning input for single-cell generation, though related text-conditioned approaches for cell-level analysis have been explored [48,49]. In parallel, contrastive vision–language models such as CLIP [14] have demonstrated that aligning two modalities into a shared embedding space enables powerful zero-shot transfer, with SimCLR [15] and MoCo [16] establishing the self-supervised contrastive paradigm and domain-specific variants such as BiomedCLIP [17] and SigLIP [18] extending it to biomedical data. Meanwhile, Diffusion Transformers (DiTs) [19] have emerged as state-of-the-art conditional generators, building on denoising diffusion probabilistic models [20,21] and latent diffusion models [22–24], and flow matching [25,50,53] provides a simulation-free training objective that simplifies the diffusion framework. The Transformer architecture [26] underpins both the DiT generator and the frozen encoders. Bridging these two lines of work—contrastive language–omics alignment and conditional diffusion generation—is the central contribution of this paper.

We present CLOP-DiT, a two-stage pipeline that combines **C**ontrastive **L**anguage–**O**mics **P**retraining (CLOP) with a conditional **Di**ffusion **T**ransformer (DiT) to generate single-cell gene expression profiles from text descriptions. The pipeline (Figure 1) first aligns text and cell embeddings into a shared contrastive space, then generates latents via flow matching, and finally decodes to gene expression. Building on BiomedBERT [27] and scGPT [28], CLOP-DiT is, to our knowledge, the first method to combine contrastive language–omics alignment with a diffusion transformer for text-conditioned single-cell generation. Across 100 evaluated cell types, CLOP-DiT achieves a KNN classification accuracy of 36.9% (well above random chance, but 52 points below the 89% real-data ceiling), a steering accuracy of 81.0%, and a diversity ratio of 0.93, indicating that text conditioning drives detectable type-specific generation, albeit with a substantial fidelity gap from real data.

The principal contributions of this work are: (1) a contrastive alignment module (CLOP) that maps text and cell embeddings into a shared space via PrototypeSigLIP loss, amplifying inter-group cosine separation from 0.006 to 0.778; (2) a 1D Diffusion Transformer (DiT1D, 22.1M parameters) that performs conditional flow matching in the 512-dimensional scGPT latent space; (3) a comprehensive four-axis evaluation framework—KNN fidelity, steering accuracy, diversity ratio, and downstream biological validation—applied to 100 cell types with bootstrap confidence intervals; and (4) an expression-level biological validation pipeline demonstrating high decoder-reconstructed gene-mean fidelity (Pearson $r > 0.999$) across five tissue contexts (four in-distribution, one study-level held-out sharing training tissue types), while revealing that per-gene variance is not preserved ($r_{\text{var}} \approx 0$).

The remainder of this paper is organized as follows: Sections 2.1–2.5 describe the dataset, model architecture, and evaluation framework; Section 3 presents training dynamics, generation quality, and downstream validation results; Section 4 discusses interpretation, limitations, and applications; and Section 5 summarizes the contributions.

2. Materials and Methods

Figure 1 provides an overview of the pipeline. We organize the Materials and Methods in three parts: (1) Dataset curation and preprocessing (2.1); (2) Model architecture and training—covering CLOP alignment (2.2), DiT generation (2.3), and scGPT decoding (2.4); (3) Evaluation framework and metrics (2.5). Together, these components form a reproducible pipeline for text-conditioned single-cell generation.

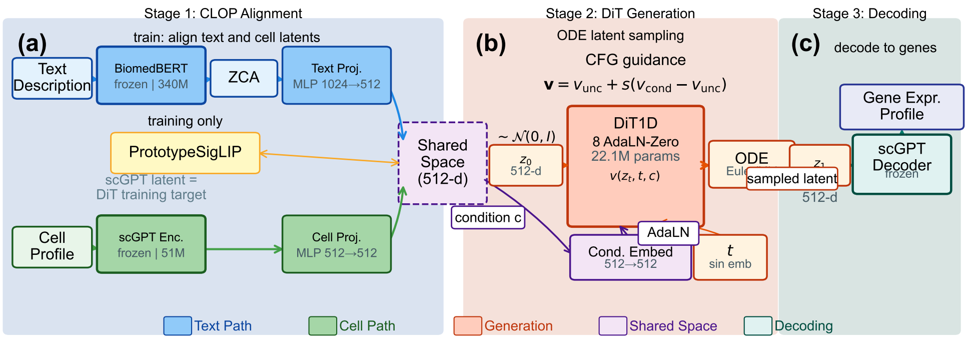

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)**: A frozen BiomedBERT-large encoder (340M parameters, 1024-d) and a frozen scGPT encoder (51M parameters, 512-d) produce text and cell embeddings, respectively. Dual MLP projectors map both modalities to a shared 512-dimensional L_2-normalized space, trained with a PrototypeSigLIP contrastive loss. ZCA whitening is applied to BiomedBERT embeddings prior to projection. **Stage 2 (DiT)**: A 1D Diffusion Transformer (8 AdaLN-Zero blocks, 22.1M parameters) performs conditional flow matching in the 512-dimensional scGPT latent space, using CLOP-projected text as the conditioning signal. An Euler or Midpoint ODE solver integrates the learned velocity field from Gaussian noise $z_0 \sim \mathcal{N}(0, I)$ to a generated latent z_1. Classifier-free guidance (CFG) is applied at inference. **Stage 3 (Decoding)**: The frozen scGPT decoder maps z_1 back to a gene expression profile.

2.1. Dataset

We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection [29,52] (2,000 highly variable genes per dataset, selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured text caption (not free-form natural language) incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template: “`{cell type}, tissue: {tissue}, organism: {organism}, markers: {top 5 DE marker genes}, context: {disease/condition}`”, where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test [57] on the training set. The exact template and marker gene sources are available in `src/data_pipeline/subcluster_annotati` (caption generation) and `scripts/data_prep/02b_enrich_descriptions.py` (evidence enrichment).

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits, and were chosen to span the major tissue and cancer categories in the corpus (lung, gastric, skin, liver, blood) via stratified selection across tissue types; the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2. Within the 72 training datasets, a cell-type-stratified split reserves ∼10% of cells per type for training-time validation, ensuring that all 69 deduplicated cell types are represented in both partitions. This held-out design tests interpolation within the training distribution; the 8 held-out datasets share tissue types and cell populations with the training corpus and do not represent a stringent out-of-distribution generalization test.

The relationship between the 1,088 text groups and the 100 evaluated cell types is as follows. The 1,088 text groups correspond to sub-cluster-level descriptions: each Leiden [30] cluster within each dataset receives a unique text caption. Many sub-clusters share the same canonical cell type (e.g., "CD8+ T cells" in lung and liver datasets produce separate text groups). For evaluation, text groups are consolidated by core cell-type identity, retaining only groups with at least 20 cells (to ensure stable centroid estimation), yielding 100 evaluable cell types. The 100-type, 200-cells-per-group design follows standard single-cell generative evaluation practice [5,7]; the minimum threshold of 20 real cells per type ensures stable centroid estimation. KNN accuracy is therefore computed over these 100 consolidated types (random chance = 1/100 = 1.0%). The text–cell alignment heatmap (Figure 5 and 6) uses 69 centroids because it operates on the deduplicated condition space, which merges cross-dataset sub-clusters into canonical types.

The 80 datasets comprise both *Homo sapiens* (∼59 datasets) and *Mus musculus* (∼21 datasets) [31], predominantly from tumor microenvironment and developmental contexts. Consequently, all generation and evaluation results should be interpreted as applying to human and mouse cancer and developmental single-cell biology; generalization to other organisms, non-cancer tissues, or perturbation conditions has not been tested.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12.0), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop.yaml`.

Hardware and compute budget.

All training and evaluation were conducted on a single NVIDIA GeForce RTX 5090 Laptop GPU (24 GB VRAM) using PyTorch 2.1 with CUDA 12.0 and automatic mixed precision (AMP/FP16). Note that this GPU was newly released at the time of submission (early 2026); the modest compute requirements (<2 GPU-hours total, <24 GB VRAM peak) mean that any Ampere-or-later GPU with ≥16 GB VRAM should suffice for reproduction. CLOP alignment training (60 epochs, batch size 1024) completes in approximately 10 minutes (∼10 s/epoch). DiT flow-matching training (200 epochs, batch size 1024) completes in approximately 75 minutes (∼23 s/epoch). End-to-end figure reproduction from cached embeddings (`scripts/pipeline/regenerate_report.sh`) takes under 30 minutes. Total compute for a single full training run (CLOP + DiT + evaluation + visualization) is under 2 GPU-hours; no multi-GPU or distributed training was used. Inference for 100 cell types (∼6,900 generated cells) takes approximately 2 minutes with 10-step Euler integration.

2.2. CLOP: Contrastive Language–Omics Pretraining

CLOP aligns text descriptions and cell profiles into a shared 512-dimensional embedding space using a PrototypeSigLIP contrastive loss, producing the well-separated condition space required by the DiT. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: $1024 \to 1024 \to 1024 \to 512$; cell: $512 \to 512 \to 512 \to 512$) with BatchNorm [32], GELU activation [33], and dropout [54] (0.2) project both modalities onto the L_2-normalized unit hypersphere. Prior to projection, Zero-phase Component Analysis (ZCA) whitening is applied to the BiomedBERT embeddings to decorrelate dimensions and counteract the known embedding-space collapse of frozen BERT models (pairwise cosine similarity > 0.88 before whitening). The ZCA transform is computed on the training set and applied identically to validation embeddings.

The CLOP loss is:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{T}\mathbf{C}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{C}\mathbf{T}^\top}{\tau}, \text{arange}(B)\right)\right] \tag{1}$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B\times 512}$ are L_2-normalized projections, $\tau = 14.0$ is a fixed logit-scale parameter (following the CLIP convention where τ multiplies rather than divides the similarity matrix, so the effective softmax temperature is $1/\tau \approx 0.071$), and label smoothing $\epsilon = 0.1$ is applied. The loss function follows the PrototypeSigLIP formulation, which aligns text prototypes to cell-group centroids rather than individual cells, with a cohesion regularization weight of 0.15. The CLOP aligner has 3.02M trainable parameters and is trained for 60 epochs with AdamW [34] (lr $= 5 \times 10^{-4}$, weight decay 0.06, cosine schedule [55] with 5-epoch warmup).

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of $\sim$130$\times$. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 8-head self-attention (head dimension 64) and feed-forward layers ($512 \to 2048 \to 512$). Timestep information is encoded via sinusoidal embeddings projected to 512 dimensions, and CLOP condition vectors are projected from 512 to 512 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ modulates each block through 6 AdaLN-Zero parameters ($\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$), all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [25,35]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \tag{2}$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution (μ=0, σ=1), which concentrates training on intermediate timesteps where the velocity field is most complex [47]. During training, classifier-free guidance (CFG) [36] is enabled by dropping the condition with 15% probability. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \tag{3}$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA [37] (decay = 0.9999).

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2.4. Decoding via scGPT

Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are decoded back to gene expression profiles using the frozen scGPT decoder. Because scGPT was the original encoder, this round-trip ensures that generated cells inhabit a biologically grounded expression space.

2.5. Evaluation Framework

We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. Random chance is 1/100 = 1.0%.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values <1 indicate mode sharpening; values >1 indicate excess noise.
4. **Downstream biological concordance**: Clustering alignment (ARI [58], NMI [59]), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 15d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

The biological evidence is organized by figure family: Figures 7–9 establish marker-level and gene-level fidelity; Figures 11–12 and 13 provide robustness readouts for guidance and diversity; Figures 14–15 report benchmarking; Figures 16 and 17 test downstream biology. Having established the methodology and evaluation framework, we now present the empirical results.

2.6. Statistical Analysis

Primary endpoints. Two operating points were pre-specified as primary comparisons: CFG = 2.0 with 10-step Euler integration (high-fidelity regime, selected because KNN accuracy plateaus at CFG 2.0–3.0 in Table A2; the lower end was chosen to minimize diversity loss) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime, selected to match DivR $\approx$ 1.0). The primary metrics are KNN top-1 classification accuracy (100-class, k=15, cosine metric on PCA-50 features) and steering accuracy (pairwise directional test, 1000 random pairs).

Exploratory analyses. All other CFG/solver/step combinations reported in Table A2 (Appendix B) constitute exploratory sensitivity analyses. No correction for multiple comparisons (e.g., Bonferroni, Holm) was applied to these comparisons. Rankings and performance claims refer only to the two primary operating points unless explicitly stated otherwise.

Uncertainty quantification. Bootstrap 95% confidence intervals (percentile method, B=1000 resamples, seed 42) are reported for all headline metrics where per-unit (per-cell or per-type) data are available (see Results). These CIs capture evaluation sampling variability but not training-run variance, as all results derive from a single training run (seed 42). Formal superiority tests (e.g., permutation tests) were not conducted; all comparisons are descriptive.

Limitations of the statistical design. (1) No multi-seed replication: inter-run variance from stochastic initialization, data shuffling, and mini-batch sampling is uncharacterized. (2) No formal pre-registration: operating points were selected based on interpretable criteria post-hoc rather than by pre-registered protocol. (3) No multiple-comparison correction for the exploratory CFG sweep.

2.7. Composite Score Formulation

The composite benchmark score aggregates all active metrics into a single ranking index. For each metric m with direction $d_m \in \{\text{lower}, \text{higher}\}$, let $v_{m,j}$ denote the raw value for method j. We compute a min–max normalised score:

$$s_{m,j} = \begin{cases} 1 - \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{lower}, \\ \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{higher}, \end{cases} \quad (4)$$

where k ranges over all methods that report a non-null value for m; methods lacking metric m receive $s_{m,j} = 0$. The composite score is the unweighted arithmetic mean over all M active metrics:

$$C_j = \frac{1}{M} \sum_{m=1}^{M} s_{m,j}. \quad (5)$$

Because not all methods support downstream biology (Section 2.5, axis 4), they receive $s = 0$ on those metrics, which inflates the score of methods that do. In the current benchmark, 8 of 17 active metrics are reported only by CLOP-DiT; the remaining 9 distributional metrics are shared by all methods. To enable transparent like-for-like comparison, we report both the full composite (C_j over all $M = 17$ metrics) and a common-metrics-only composite (C_j^{common} restricted to the 9 shared metrics). Readers should interpret the full composite as a combined capability ranking rather than a fair distributional quality comparison. Table 1 and the per-metric panels in Figure 15 report individual metrics for transparent comparison. Bootstrap 95% confidence intervals (percentile method, B=1,000) for all headline generation metrics are provided in the supplementary materials (Table S5).

3. Results

We report in-distribution results along the four evaluation axes defined in the Methods, organized by evidence type: training dynamics and embedding space (Figures 2–3); core generation quality metrics (Figures 4–5 and 6); gene-level fidelity (Figures 7–9); conditioning robustness and diversity trade-offs (Figures 10–12 and 13); baseline comparisons and benchmarking (Figures 14–15); and downstream biological validation (Figures 16–17). This structure allows readers to assess evidence quality at multiple resolutions, from training convergence through partial downstream biological utility.

3.1. Training Dynamics

Figure 2 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 60, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%)—a 256-way contrastive ranking task in which the model must match each text to its corresponding cell from a full mini-batch—and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a final validation velocity cosine of 0.990 with EMA, indicating near-perfect velocity field prediction. Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 → 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

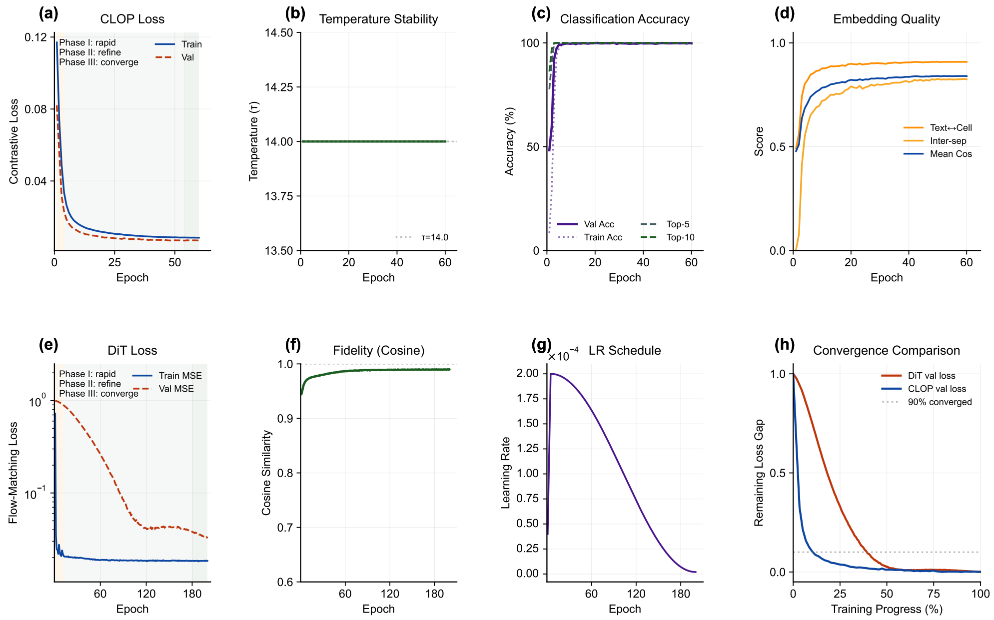

Figure 2. Training dynamics of the CLOP-DiT pipeline. (**a**) CLOP contrastive loss, (**b**) learned logit-scale parameter (final $\tau = 14.0$, corresponding to effective softmax temperature $1/\tau \approx 0.071$), (**c**) batch classification accuracy, and (**d**) embedding quality metrics over 60 epochs. Despite modest validation accuracy (∼2.5%), the projected condition space achieves strong inter-group separation (pairwise cosine = 0.222). (**e**) DiT train and validation MSE loss (log scale), (**f**) velocity cosine similarity, (**g**) learning rate schedule, and (**h**) normalized convergence comparison of CLOP and DiT validation losses, plotted against training progress (%); the dashed line marks the 90% convergence threshold. Final metrics: CLOP val loss = 8.5×10^{-3}, prototype accuracy = 99.8%, 60 epochs; DiT val MSE = 3.2×10^{-2}, val cosine = 0.990, final LR = 2.0×10^{-6}, EMA decay = 0.9999, 10-step Euler sampler, 200 epochs.

3.2. CLOP Embedding Space

Figure 3 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP [38]: text and cell embeddings jointly projected into the shared 512-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

3.3. Core Evaluation Metrics

Table 1 and Figure 4 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (37× random chance; 52 points below the 89% real-data baseline), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Note that the Fréchet distance (FD) reported in Table 1 is often misleading for conditional generation, because global mean/covariance matching tends to favour unconditional baselines; FD should therefore be interpreted alongside the directional metrics (KNN, steering, diversity). Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering =

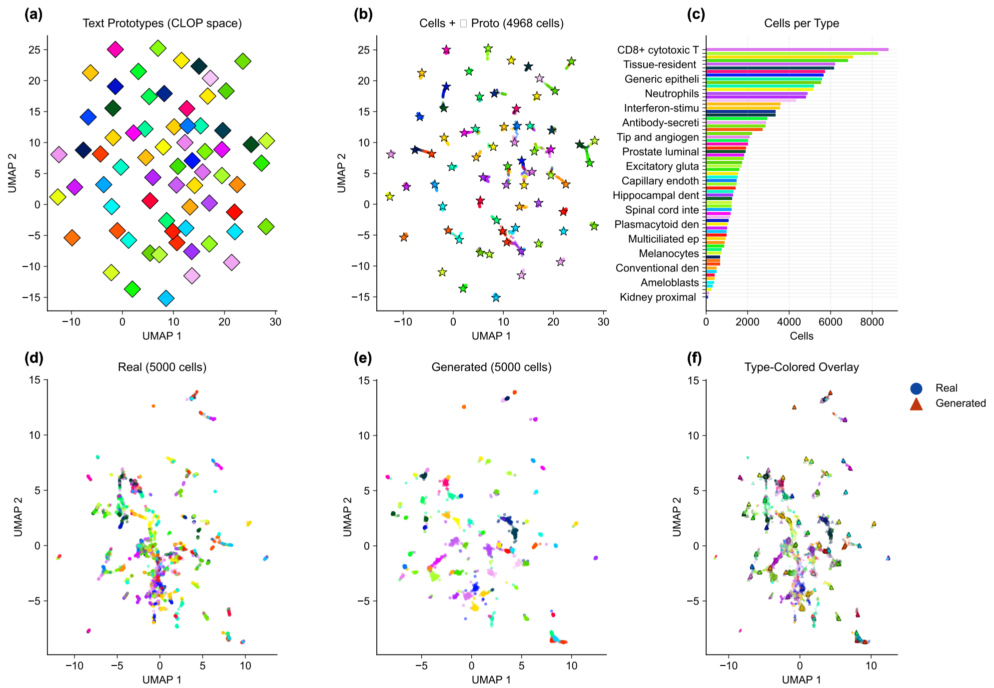

Figure 3. Embedding space analysis. (**a**) UMAP of text prototypes in the shared CLOP projection space, (**b**) cell embeddings with text prototype overlay (star markers) confirming co-localization of text and cell clusters (pairwise cosine = 0.222), and (**c**) cells-per-type bar chart with abbreviated type labels. (**d**) UMAP of real cells only (type-colored), (**e**) UMAP of generated cells only (type-colored), and (**f**) type-colored overlay of real (circles) and generated (triangles) cells. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts.

47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Table 1. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 k-nearest-neighbor accuracy among 100 cell types (random = 1.0%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy in embedding space (trained on real, evaluated on generated; distinct from the expression-space downstream classifier in Figure 16). FD: Fréchet distance [39] (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

Bootstrap confidence intervals. To quantify evaluation sampling uncertainty, we computed bootstrap 95% CIs by resampling the 69 cell types present in the generation cache with replacement (B=1,000, percentile method); the remaining 31 of the 100 atlas types lack sufficient generated cells for bootstrap estimation. At an intermediate reference configuration (CFG = 1.5, Midpoint, 20 steps; selected to balance fidelity and diversity between the two primary operating points): KNN top-1 = 0.509 [0.421, 0.586], KNN top-5 = 0.728 [0.658, 0.794], steering = 1.000 [1.000, 1.000] (ceiling effect: all bootstrap resamples achieve perfect pairwise accuracy at this configuration), diversity ratio = 0.969 [0.963, 0.975], linear accuracy = 0.583 [0.518, 0.646], and centroid cosine = 0.898 [0.875, 0.913]. The narrow

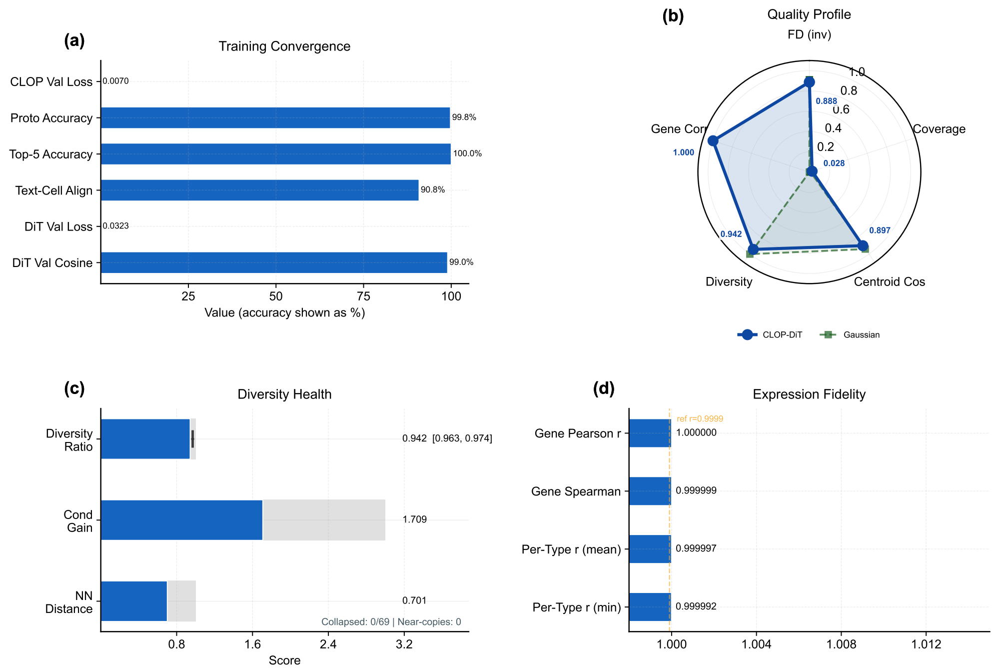

Figure 4. Metrics dashboard summarizing conditional generation quality. (**a**) Training convergence: normalized endpoint scores for loss reduction and accuracy/correlation with exact final values beside each row. (**b**) Quality profile radar chart showing multi-metric comparison across generation configurations. (**c**) Diversity health gauges. (**d**) Expression fidelity bar chart. KNN accuracy (well above chance), steering accuracy (81%), diversity ratio, linear classifier accuracy, and Fréchet distance are shown across configurations. The collapse of unconditional generation to random baselines validates that text conditioning drives all cell-type specificity.

diversity ratio and centroid cosine CIs reflect uniformly high performance across types; the wider KNN CIs reflect type-dependent classification difficulty. These CIs capture evaluation sampling variability only, not training-run variance (single seed; see Section 4).

3.4. Per-Type Generation Quality and Text–Cell Alignment

Beyond aggregate metrics, per-type analysis reveals which cell types benefit most from text conditioning and where failures concentrate. Figure 5 presents per-type generation quality. The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across all 100 atlas cell types; the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as color. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. Figure 6 shows the text–cell alignment analysis, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

3.5. Marker Gene Fidelity

A key question for conditional generation is whether cell-type-specific marker gene signatures are preserved. Figure 7 compares the expression of canonical marker genes between real and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific expression signatures. This directly shows that the biological signal resides in the intended marker genes rather than only in latent-space summary statistics.

3.6. Expression Correlation and Dimension Analysis

To assess gene-level fidelity beyond marker genes, Figures 8 and 9 quantify the per-gene Pearson correlation between real and generated mean expression across cell types and provide a broader view of expression distribution properties, including norm matching

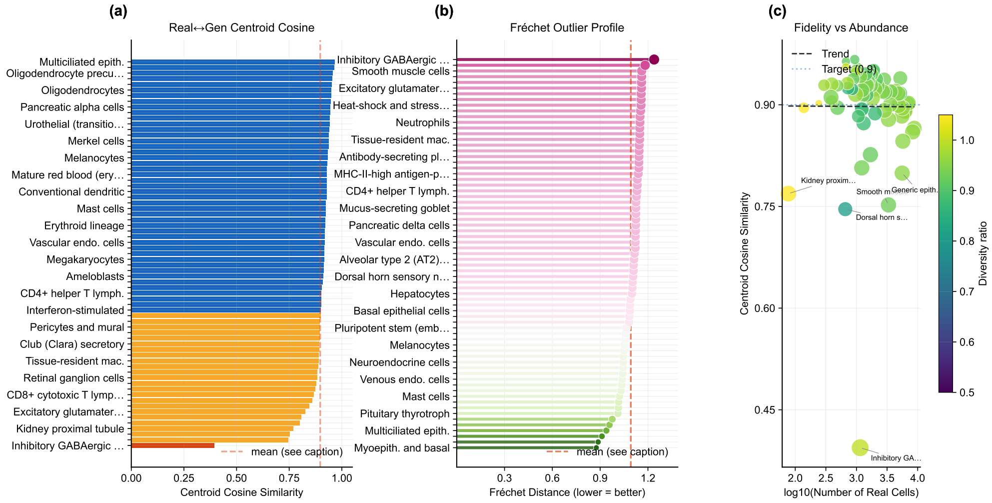

Figure 5. Per-type generation fidelity. (**a**) Per-type centroid cosine similarity ranked across cell types, (**b**) Fréchet outlier profile with the global mean marked, and (**c**) abundance–fidelity scatter (bubble size ∝ Fréchet distance, color encodes diversity ratio). Cell-type names are mapped by rank position in the supplementary materials.

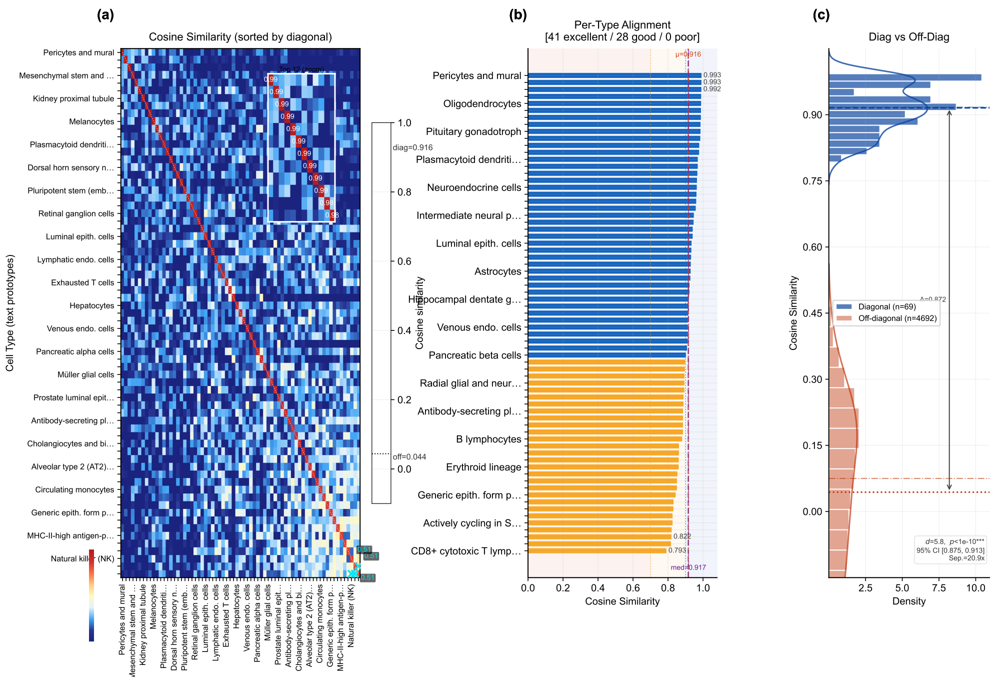

Figure 6. Text–cell alignment. (**a**) 69×69 cosine similarity heatmap between text and cell centroids, sorted by diagonal similarity. Mean diagonal similarity $\mu_{\text{diag}} = 0.916$ ($\sigma = 0.053$), mean off-diagonal $\mu_{\text{off}} = 0.044$ ($\sigma = 0.205$); Cohen's $d = 5.8$, separation ratio 20.9×, Mann–Whitney U $p < 10^{-10}$. (**b**) Per-type alignment bars color-coded by quality tier (41 excellent ≥ 0.9, 28 good, 0 poor). (**c**) Diagonal vs. off-diagonal distribution comparison with KDE overlay; the two populations are well separated ($\Delta = 0.872$, bootstrap 95% CI [0.875, 0.913]).

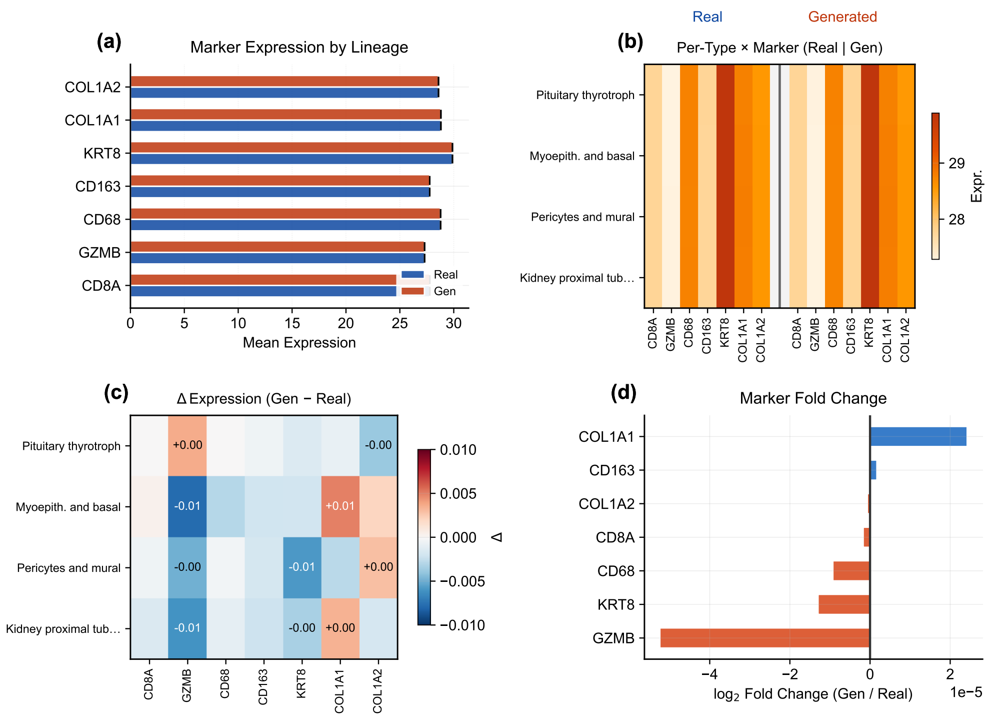

Figure 7. Marker gene comparison. (**a**) Paired bar chart of canonical marker gene expression in real (solid) vs. generated (hatched) cells, with bar colors grouping markers by lineage family (CD8 T, myeloid, epithelial, stromal); fold-change annotations flag markers deviating from unity. (**b**) Per-type × marker dual heatmap showing real and generated expression side by side for selected cell types. (**c**) Difference heatmap (Δ = generated − real) with cell annotations for large deviations. (**d**) Fold-change waterfall for all markers, with a shaded band marking the ±5% agreement zone. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

(generated mean norm 20.97 vs. real 20.89) and per-dimension statistics. Together they localize biological meaning in gene-wise means, variances, and marker-level residuals rather than only in embedding geometry.

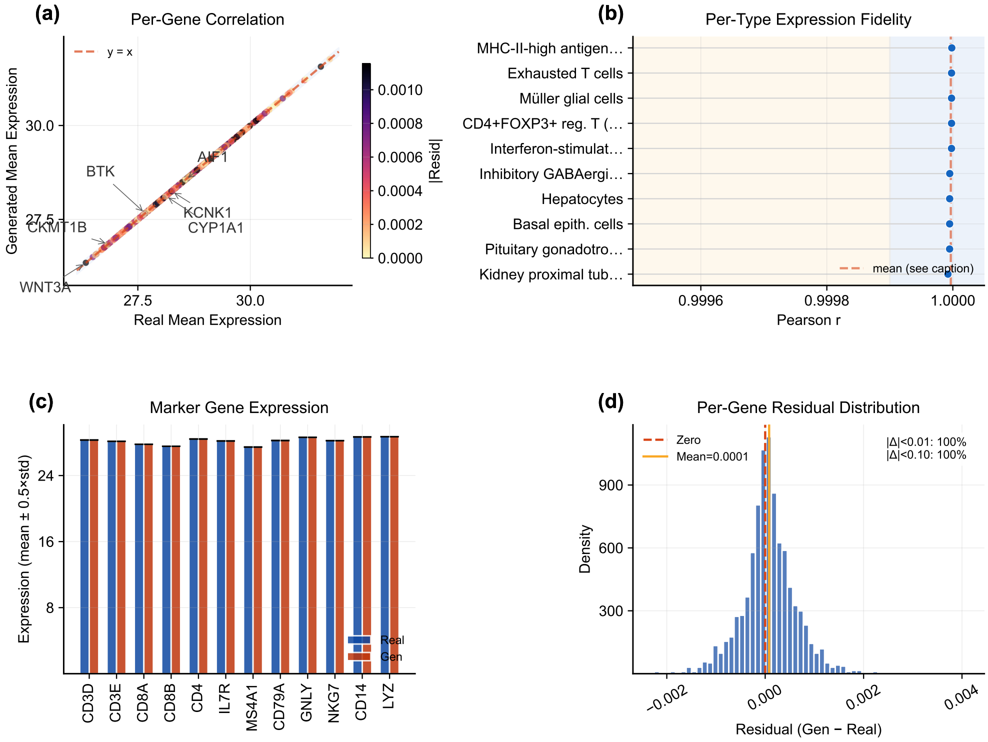

Figure 8. Expression correlation analysis. (**a**) Per-gene mean expression scatter (real vs. generated), with point color encoding absolute residual magnitude and outlier genes annotated. (**b**) Per-type Pearson r lollipop chart with threshold-band shading. (**c**) Marker gene expression (real vs. generated grouped bars with error whiskers). (**d**) Per-gene residual distribution (generated − real) with percentage of genes within tolerance bands. High correlation confirms that the scGPT decoder faithfully reconstructs gene-level signatures from DiT-generated latent vectors.

3.7. Conditioning Landscape and Diversity Trade-Offs

Figure 10 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance. Together with Figures 11, 12, and 13, this section is the main robustness analysis for whether control can be increased without erasing biologically meaningful heterogeneity.

Building on this conditioning structure, a sweep across CFG $\in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 11). KNN accuracy saturates near CFG = 2.0–3.0 (41$\times$ random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.930) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

Figure 12 and Figure 13 further quantify the two operating regimes: high-fidelity (CFG $\geq$ 2.0, DivR $\approx$ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augment the global trade-off curves with per-type diversity-tail diagnostics. Expression-level diversity measurements confirm that the Midpoint solver best preserves gene-level variance.

3.8. Baseline Comparison and Composite Benchmark

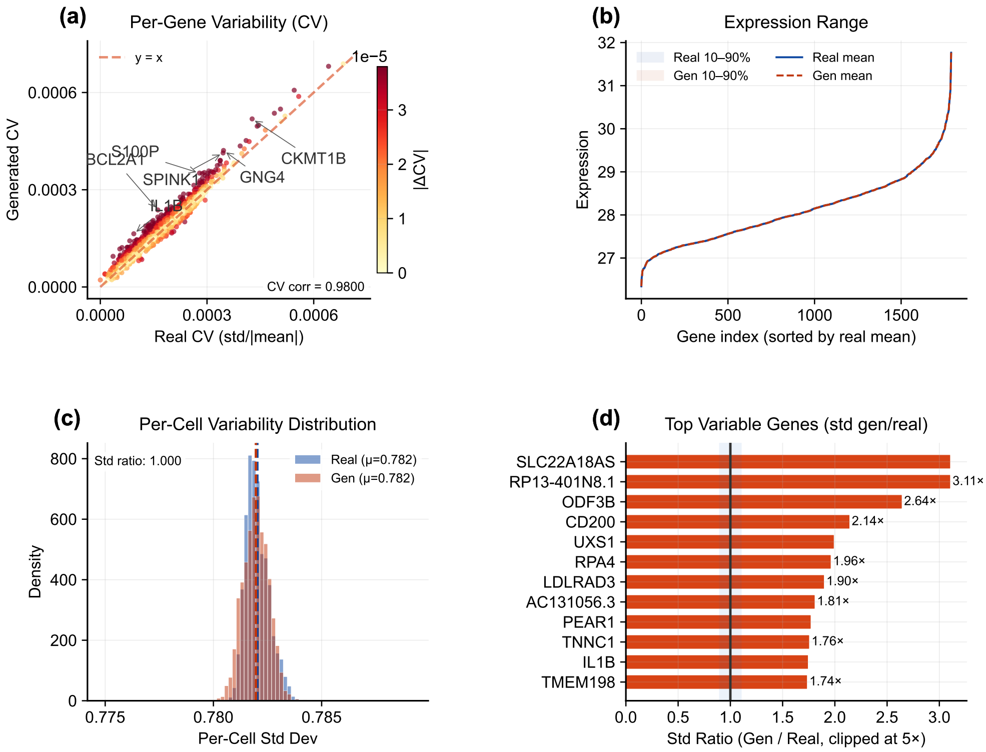

Figure 9. Expression distribution analysis. (**a**) Per-gene coefficient of variation scatter (real vs. generated) with outlier gene annotations. (**b**) Expression range ribbon with 10–90th and 25–75th percentile bands for real and generated cells. (**c**) Per-cell variability (standard deviation) distribution comparing real and generated populations. (**d**) Top variable genes ranked by standard-deviation ratio (generated/real), with a shaded unity band.

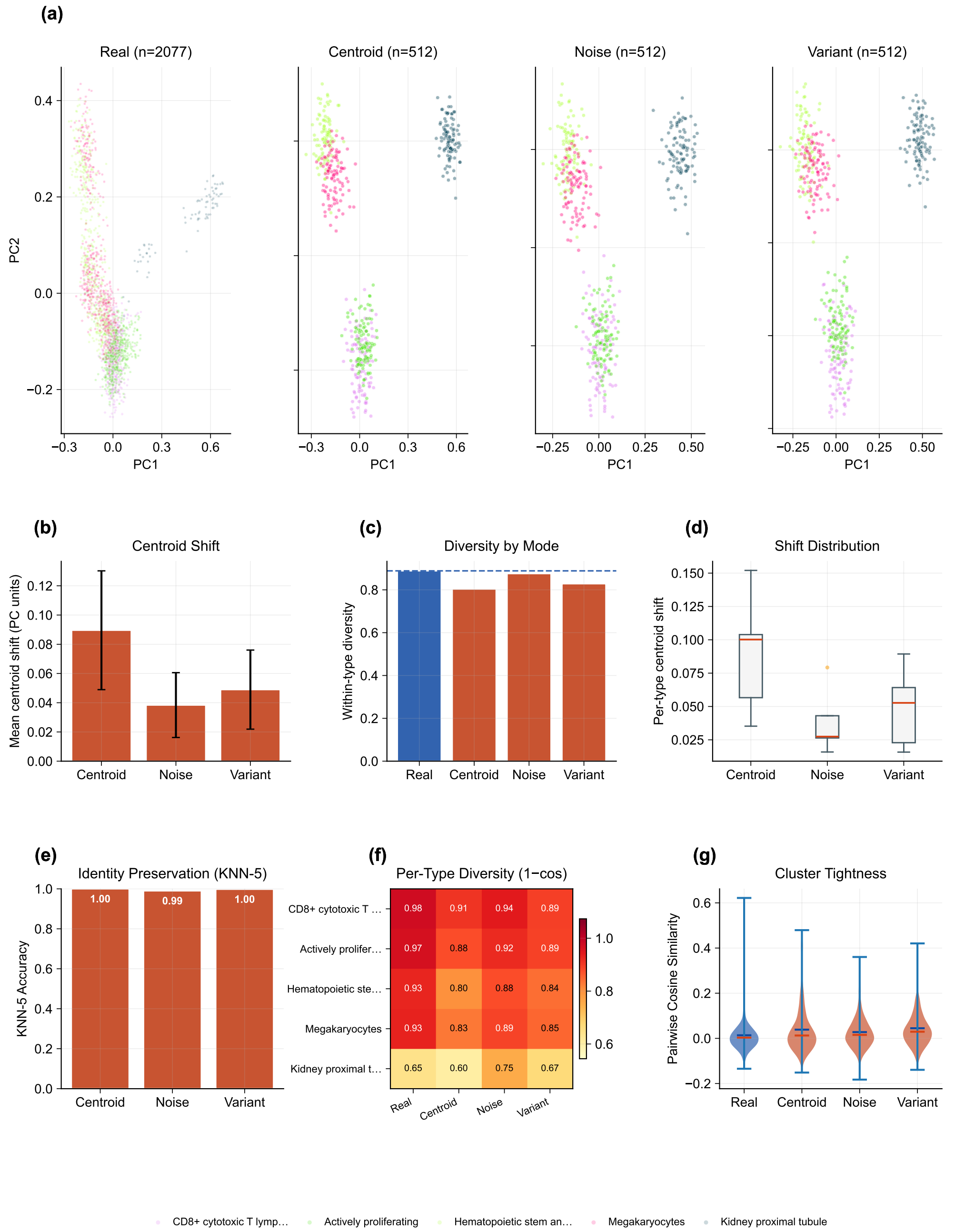

Figure 10. Conditioning landscape. (**a–d**) PCA projections of the CLOP condition space for real cells and three conditioning modes (centroid, noisy, variant), showing cell counts and mean within-type diversity per mode. (**e**) Mean centroid shift from each generated mode to the real reference. (**f**) Within-type diversity by conditioning mode (dashed line = real baseline). (**g**) Per-type centroid shift distributions (box plots). (**h**) KNN-5 identity-preservation accuracy per conditioning mode. (**i**) Per-type diversity heatmap (1 − mean cosine). (**j**) Pairwise cosine violin plots comparing cluster tightness across modes.

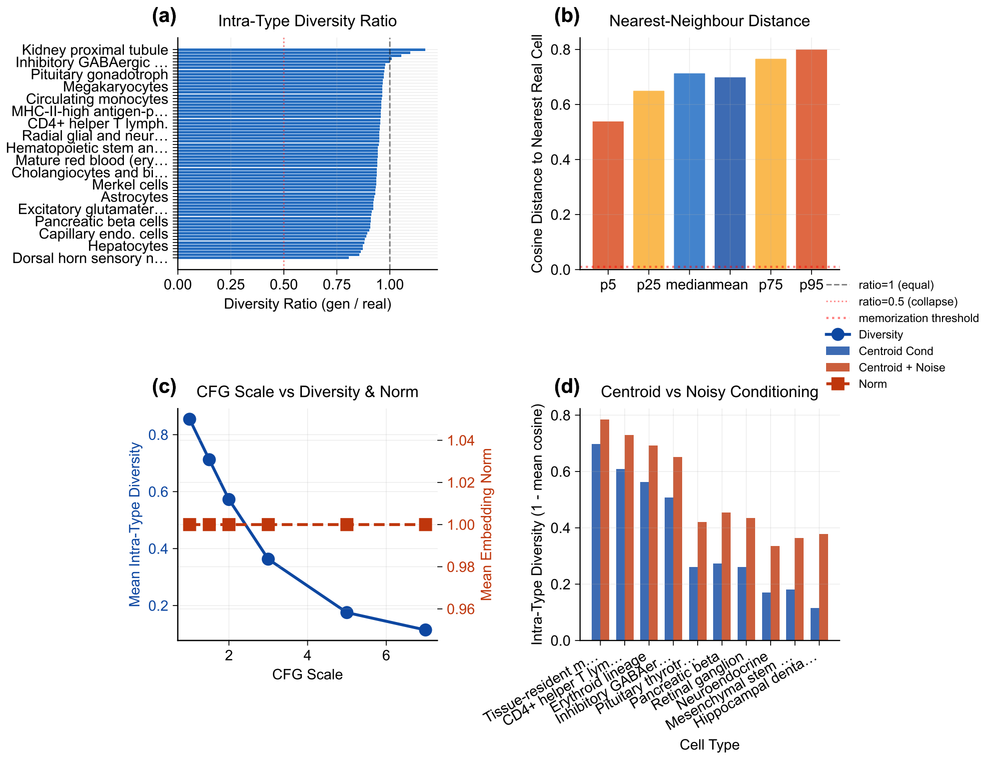

Figure 11. Diversity diagnostics. (**a**) Intra-type diversity ratio ranked across cell types (labels abbreviated for readability). (**b**) Nearest-neighbor distance distributions for real and generated populations. (**c**) CFG scale vs. diversity and norm trade-off across Euler and Midpoint solvers. (**d**) Centroid vs. noisy conditioning comparison. KNN accuracy saturates at CFG $\approx$ 2.0–3.0 while diversity decreases monotonically.

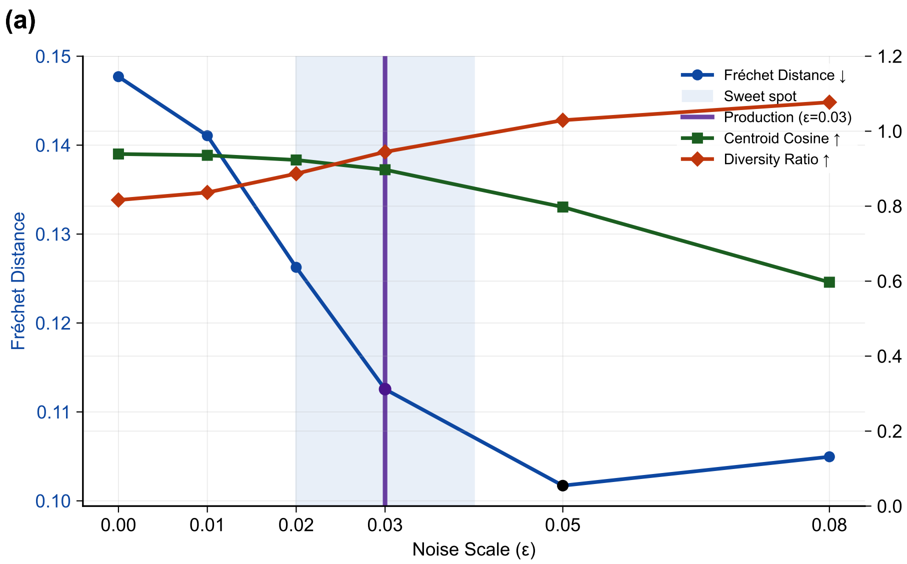

Figure 12. Noise–scale trade-off. (**a**) Fréchet distance (left axis), centroid cosine similarity, and diversity ratio (right axis) as a function of noise scale ε at CFG = 1.5. The production configuration ($\varepsilon = 0.03$) is highlighted; a shaded band marks the sweet-spot region.

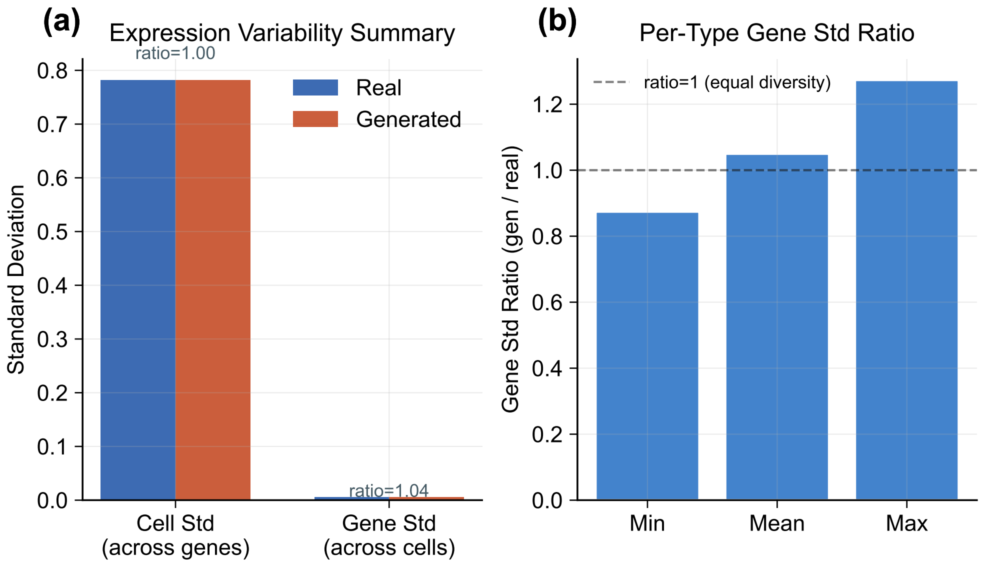

Figure 13. Expression-level diversity. (**a**) Gene-expression variability summary comparing real and generated populations across key statistics. (**b**) Per-type gene standard-deviation ratio (generated/real), confirming that the recommended Midpoint solver preserves biologically meaningful variance.

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, shuffled-label and mean-collapse controls in the benchmarking pipeline, and decoder-only ablations. Figure 14 shows that CLOP-DiT is strongest on conditioning-sensitive and downstream-relevant metrics (e.g., steering and classifier transfer), while some unconditional distribution-matching baselines may outperform on FD-like mean-matching objectives (see Discussion for the Fréchet distance caveat).

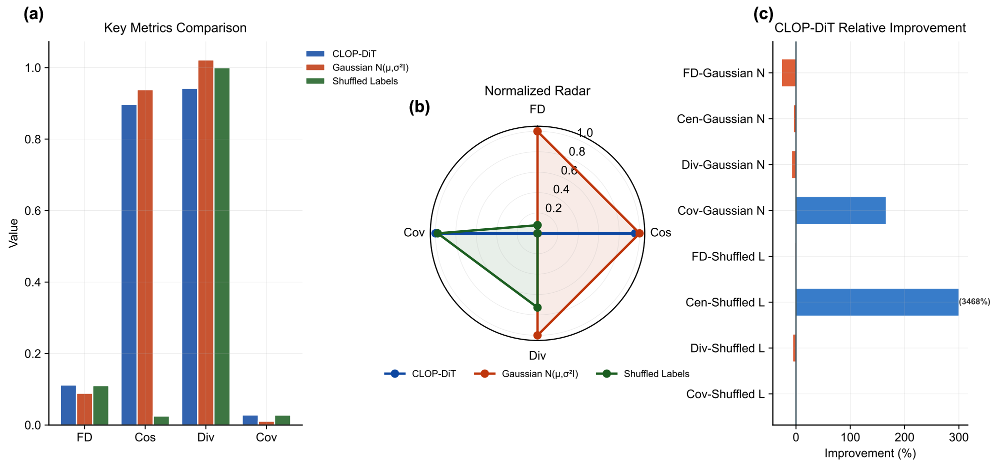

Figure 14. Baseline comparison. (**a**) Head-to-head grouped-bar analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage). (**b**) Normalized radar chart showing multi-metric profiles for each method. (**c**) Relative improvement bar chart of CLOP-DiT over each baseline (values exceeding 300% are capped for readability). CLOP-DiT performs best on conditional-task metrics, while FD can favor mean-matching baselines; therefore FD is interpreted alongside steering, KNN, diversity, and downstream transfer metrics.

Figure 15 aggregates these metrics into a composite benchmark score. CLOP-DiT achieves a full composite score of 0.842 across all 17 metrics (9 distributional + 8 downstream biology), ranking first among the seven methods evaluated in our benchmark suite (including the newly-added scVI Latent baseline; see Methods, Equations 4–5, for the composite formula and sensitivity analysis). However, when restricted to the 9 distributional metrics shared by all methods, the Gaussian baseline (0.750) outperforms CLOP-DiT (0.702), revealing that the full-composite advantage is driven by the 8 CLOP-DiT-exclusive downstream metrics. The scVI Latent baseline achieves the highest coverage and density but ranks lower on centroid cosine and collapsed-type metrics. Bootstrap 95% confidence intervals support the stability of per-metric orderings but cannot overcome the fundamental compositional bias of the full composite.

3.9. Downstream Biological Validation

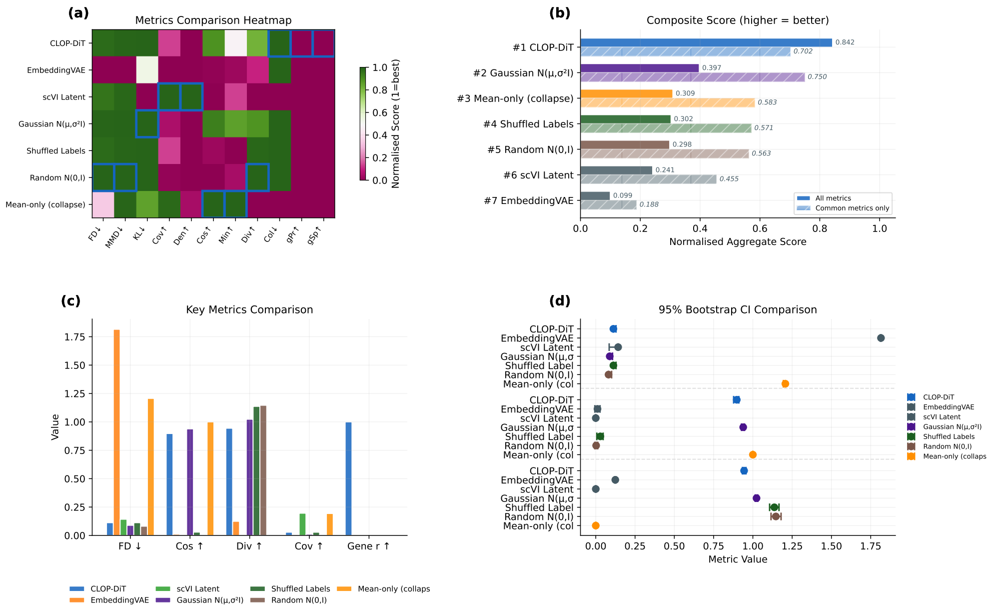

Figure 15. Composite benchmark. (**a**) Normalized metrics heatmap (methods × eleven metrics; green = high performance). (**b**) Composite score: solid bars show the full 17-metric composite, hatched bars show the common-metrics-only (9 shared distributional metrics) composite. (**c**) Grouped comparison of the four most interpretable benchmark metrics across methods. (**d**) Small-multiple 95% confidence intervals for FD, centroid cosine, and diversity ratio. On the full composite (0.842) CLOP-DiT ranks first, but on the common-metrics-only composite (0.702) it is outperformed by the Gaussian baseline (0.750), indicating that the overall advantage is driven by CLOP-DiT-exclusive downstream metrics.

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. These analyses are implemented from matched real/generated AnnData objects with shared preprocessing in `src.evaluation.downstream_biolog` using Scanpy [40] for single-cell analysis, ensuring that any downstream advantage is not caused by mismatched feature engineering. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 16): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden [30] clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores [41]. High mixing scores indicate that generated cells partially co-localize with real cells of the same type in the joint embedding, though the discriminator AUC of 0.656 confirms that real and generated populations remain distinguishable. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, demonstrating that downstream performance exhibits strong heterogeneity across cell types. The real-vs.-generated discriminator AUC (0.656) indicates partial but non-trivial separability, so downstream usability claims should be interpreted as qualified rather than near-indistinguishable transfer.

DE concordance (Figure 17): Wilcoxon rank-sum differential expression analysis [57] is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, resident macrophage vs. monocyte, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each gene by effect size and colors it by adjusted significance, which separates minor noisy disagreements from the biologically important genes that dominate each contrast.

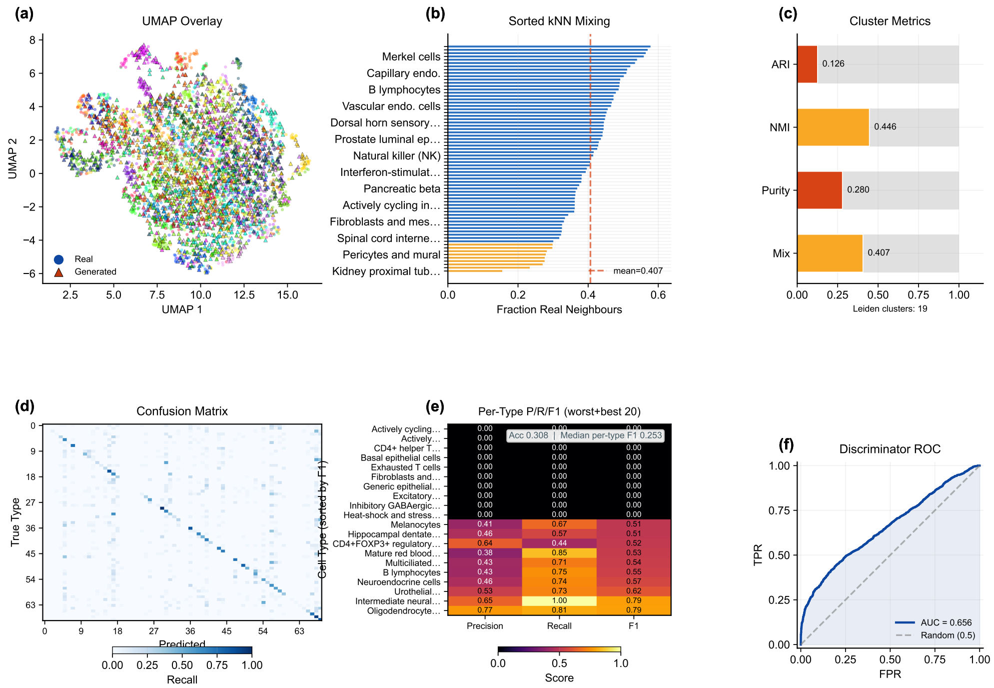

Figure 16. Downstream validation: clustering and classifier alignment. (**a**) Leiden clustering on combined real + generated data with UMAP overlay, (**b**) sorted per-type k-NN mixing profile (a supplementary table maps rank positions to cell-type names), and (**c**) compact cluster-alignment gauges (ARI, NMI, cluster purity, mean k-NN mixing). (**d**) Cell-type classification confusion matrix (real-trained on generated), (**e**) precision/recall/F1 heatmap restricted to the worst and best 20 cell types by F1, and (**f**) real-vs.-generated discriminator ROC curve. Median per-type F1 is reported in the heatmap stat box.

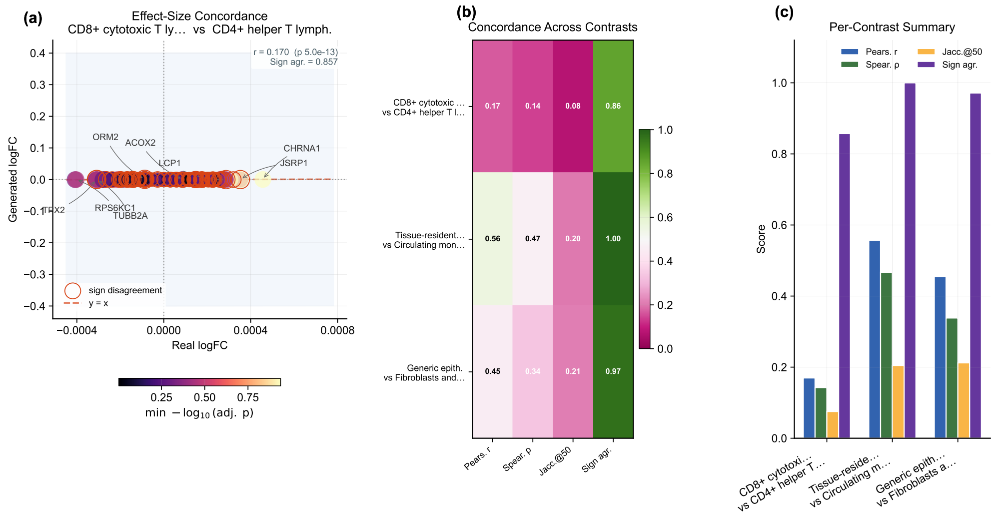

Figure 17. DE concordance. (**a**) Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and color indicating adjusted significance; sign-disagreement genes are outlined. (**b**) Compact summary heatmap reporting four metrics—Pearson r, Spearman ρ, Jaccard overlap of the top-50 DE genes (Jaccard@50), and sign agreement—across the three biologically meaningful contrasts (CD8 vs. CD4 T, resident macrophage vs. monocyte, epithelial vs. fibroblast). (**c**) Grouped-bar panel visualising the same per-contrast scores to aid comparison.

Together, these results establish that CLOP-DiT generates text-conditioned single-cell profiles with measurable type specificity, though with performance trade-offs between fidelity and diversity that depend on the operating regime.

3.10. Cross-Dataset Biological Validation

To assess biological fidelity across tissue contexts, we performed expression-level validation on five datasets spanning lung adenocarcinoma (GSE123902), basal cell carcinoma (GSE123813), liver T-cell hepatocellular carcinoma (GSE98638), gastric cancer (GSE183904), and acute lymphoblastic leukemia in bone marrow (GSE132509). Four of these five datasets (GSE123902, GSE123813, GSE98638, GSE132509) are from the training split; only GSE183904 is among the 8 held-out validation studies (Table S2). This analysis therefore primarily evaluates in-distribution gene-level reconstruction fidelity rather than out-of-distribution generalization. For each dataset, cells were generated conditioned on the dataset-specific text descriptions and compared with real (scGPT-reconstructed) expression profiles; because both the generated and reference expressions are decoded by the same frozen scGPT decoder, the resulting correlations measure decoder-reconstructed fidelity rather than agreement with raw sequencing counts. Per-gene mean expression, gene-level variance, and Fréchet distance in both gene space and embedding space were computed for each dataset independently (Table 2).

Table 2. Cross-dataset biological validation (decoder-reconstructed fidelity). Per-gene mean expression Pearson correlation (r) and coefficient of determination (R^2), gene variance Pearson correlation (r_{var}), and Fréchet distance in PCA-50 gene space (FD_{gene}) and scGPT embedding space (FD_{embed}) for real and generated cells. Both "real" and generated expression profiles are decoded by the same frozen scGPT decoder; correlations therefore measure decoder-reconstruction consistency rather than agreement with raw sequencing counts. Split indicates whether the dataset was used for training (Train) or held-out validation (Val). All five datasets achieve gene_mean $r > 0.999$.

Dataset (GEO)	Tissue	Split	r_{mean}	R^2	r_{var}	FD_{gene}	FD_{embed}
GSE123902	Lung	Train	0.9995	0.9990	$+0.011$	3.07×10^{-5}	73.9
GSE123813	Skin (BCC)	Train	0.9997	0.9995	-0.019	5.25×10^{-6}	72.7
GSE98638	Liver	Train	0.9995	0.9989	-0.003	2.04×10^{-5}	155.4
GSE183904	Gastric	Val	0.9997	0.9994	$+0.013$	1.30×10^{-5}	62.1
GSE132509	Blood (ALL)	Train	0.9993	0.9985	-0.007	8.87×10^{-6}	69.1

Across all five datasets, per-gene mean expression Pearson correlation exceeds 0.999 (range: 0.9993–0.9997), and R^2 exceeds 0.998 (range: 0.9985–0.9995). Because both reference and generated profiles pass through the same frozen scGPT decoder, these correlations reflect decoder-reconstructed fidelity: any latent vector near the training manifold decodes to a similar expression profile, so the high r quantifies round-trip decoder consistency rather than absolute biological accuracy against raw counts. Fréchet distance in PCA-50 gene space is consistently below 10^{-4} (range: 5.25×10^{-6} to 3.07×10^{-5}), indicating excellent gene-space distributional agreement at the level of first and second moments.

However, per-gene variance correlation (r_{var}) is near zero or slightly negative across all datasets (range: -0.019 to $+0.013$), indicating that while the model reproduces mean expression accurately, it does not faithfully preserve gene-level variability structure. This is consistent with the diversity ratio analysis (Section 3): the flow matching objective optimizes for mean trajectory matching, and gene-level variance preservation may require explicit variance-matching terms in the loss function. The embedding-space Fréchet distance (FD_{embed}) ranges from 62.1 to 155.4 across datasets, substantially higher than gene-space FD. This apparent discrepancy—high decoder-reconstructed gene-mean fidelity alongside high embedding-space FD—arises because the scGPT decoder maps a broad range of latent vectors to similar expression profiles (many-to-one mapping), so gene-level fidelity does not require precise latent-space distributional matching.

3.11. CLOP Ablation Study

To validate the contribution of individual CLOP components, we conducted 15 ablation experiments, removing or modifying one component at a time while holding all other settings constant. Table 3 summarizes the six most informative ablations.

Table 3. CLOP ablation study. Validation prototype accuracy (top-1) for the baseline configuration (v8.2, all features enabled) and representative ablations. Δ denotes the change from baseline in percentage points.

Ablation	Category	Val Proto Acc (%)	Δ (pp)
−cohesion	regularization	86.41	+10.24
−cell noise	regularization	86.23	+10.07
−dropout (lighter)	regularization	77.32	+1.15
Baseline (v8.2)	reference	76.17	—
Fixed temperature	optimization	71.55	−4.61
High variant prob	regularization	70.14	−6.03

The two most impactful ablations are cohesion regularization removal (+10.24 pp) and cell noise augmentation removal (+10.07 pp), both of which improve prototype accuracy. This counterintuitive result suggests that these regularization techniques, while designed to prevent overfitting, may over-regularize the small-scale contrastive objective by forcing excessively tight within-group clustering before inter-group separation has been established. Architecture ablations (projection dimension 256 vs. 512 vs. 768) have minimal impact (within ±0.3 pp), indicating that the CLOP aligner is not sensitive to projection dimension in this range. Fixed temperature (−4.61 pp) and high variant probability (−6.03 pp) degrade performance substantially, confirming that learned temperature scheduling and moderate augmentation are both beneficial. Note that prototype accuracy measures text-to-centroid alignment rather than downstream generation quality; the ablation results characterize the CLOP alignment stage specifically and do not directly predict DiT generation performance.

3.12. Rare Cell Augmentation

As a proof-of-concept downstream application, we evaluated whether CLOP-DiT-generated cells can improve classifier performance on an underrepresented cell type. We identified "Cycling cells" as the rarest type in a test dataset (below 5% prevalence), downsampled the training set to exacerbate the class imbalance, and augmented with synthetic cells at 1×, 5×, and 10× ratios. The baseline classifier (trained on real cells only) achieves an overall macro F1 of 0.964 and a Cycling-cell-specific F1 of 0.828. After augmentation at 1×, overall macro F1 remains 0.953 and Cycling-cell F1 is 0.786; at 5× and 10×, macro F1 stabilizes at 0.947 with Cycling-cell F1 at 0.741. The augmented cells therefore do not improve the rare-type classifier on this dataset, but they also do not substantially degrade overall classification, indicating that the synthetic profiles are sufficiently realistic to be mixed into training data without catastrophic interference. A more targeted augmentation strategy—selecting the most confident generated cells or combining with real-cell oversampling—may be needed to achieve a net positive augmentation effect.

4. Discussion

CLOP-DiT addresses what is, to our knowledge, a previously unmet challenge: generating single-cell gene expression profiles conditioned on structured text descriptions of cell identity. The text conditioning follows a rigid template (cell type, tissue, organism, marker genes, disease context) rather than free-form natural language; the model may therefore be learning an enriched form of label conditioning rather than general language understanding (see *Marker gene and template caveats* below). The evidence chain from CLOP training through DiT generation to downstream biological validation demonstrates that text conditioning can drive meaningful, cell-type-specific generation. Expression-level validation across five tissue contexts (Table 2; four from the training split, one study-level held-out) confirms high decoder-reconstructed gene-mean fidelity ($r > 0.999$), while the CLOP ablation study (Table 3) identifies cohesion regularization and cell noise as over-regularizing components whose removal improves alignment accuracy by over 10 percentage points.

Gene variance preservation. A consistent finding across all five cross-dataset validation experiments is that gene-level variance correlation is near zero (r_{var} range: -0.019 to $+0.013$; Table 2), despite high decoder-reconstructed mean fidelity. This indicates that the flow matching objective captures the conditional expectation of each gene accurately but does not reproduce the stochastic variability around that expectation. From a biological perspective, this means generated cell populations share the correct average expression signature but lack the cell-to-cell heterogeneity characteristic of real tissue samples. This is a critical limitation: cell-to-cell gene expression variance is the primary signal for trajectory inference, pseudotime analysis, gene regulatory network reconstruction, and identification of transition states and rare subpopulations. Generated cells are therefore effectively mean-expression representatives rather than biologically faithful single cells. Preserving this heterogeneity likely requires explicit variance-matching or distributional loss terms during training, and represents a key direction for future work.

Interpreting the KNN gap. Generated cells achieve 36.9% KNN accuracy (vs. 1.0% random chance) but fall 52 percentage points short of real-data classification levels (89.0%). This gap reflects the difficulty of the generation task: real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, meaning inter-type differences occupy only $\sim$0.6% of the embedding variance. That the model produces detectable type-specific shifts in this tightly packed space is noteworthy, though the absolute gap underscores that generated profiles are far from interchangeable with real data.

Two operating regimes. The CFG sweep reveals two complementary regimes. A high-fidelity regime (CFG = 2.0–3.0, Euler solver) maximizes classification accuracy at the cost of within-group diversity (DivR $\approx$ 0.5). A high-diversity regime (CFG = 1.0, Midpoint solver) achieves near-ideal variance preservation (DivR = 0.93) while maintaining 80.7% steering accuracy. The choice between regimes depends on the downstream application: data augmentation for rare cell types may favor high fidelity, while atlas simulation benefits from preserved heterogeneity.

Fréchet distance can be misleading for conditional generation. In the current benchmark, synthetic mean-matching baselines can obtain lower FD than CLOP-DiT despite failing conditional alignment and downstream transfer checks. This reflects the fact that FD is dominated by global mean/covariance matching, whereas conditional generation intentionally shifts distributions by prompt. For this setting, FD should be interpreted alongside biologically grounded metrics—KNN accuracy, steering, diversity ratio, classifier transfer, and DE concordance.

Frozen encoder design choice. Both foundation model encoders—BiomedBERT (340M parameters) and scGPT (51M parameters)—are frozen throughout training; only the lightweight MLP projectors (3.02M parameters combined) are updated. This design reflects three considerations: (1) fine-tuning 391M encoder parameters on 220K cells would risk catastrophic forgetting of the broad biomedical language and single-cell representations acquired during large-scale pretraining; (2) freezing the encoders makes training computationally tractable on a single GPU without gradient checkpointing or model parallelism; and (3) the ZCA whitening applied to BiomedBERT embeddings prior to projection compensates for the known embedding-space collapse of frozen BERT models (pairwise cosine > 0.88) without modifying encoder weights. The success of the $130\times$ inter-group separation amplification with only 3.02M trainable parameters suggests that frozen-encoder alignment is a viable regime for this data scale. Nonetheless, adapter-based fine-tuning (e.g., LoRA [42]) of the text encoder could improve alignment for challenging cell types and is a natural next step.

Why the biological evidence is credible. The manuscript does not rely on a single aggregate metric. Biological meaning is localized across marker-gene preservation (Figure 7), gene-level correlation and variance structure (Figures 8 and 9), per-type heterogeneity and diversity tails (Figures 5, 6, 12, and 13), and downstream reuse in clustering, classification, and DE workflows (Figures 16 and 17). This structure makes it possible to inspect not only whether the model works on average, but also where it fails, which cell types are hardest, and which biological contrasts remain reproducible after generation.

Broader biological applications. Although the present study focuses on text-conditioned generation, the evaluation framework could in principle be extended to adjacent tasks such as rare-cell augmentation, cell-state editing, and perturbation-style contrasts, though none of these have been experimentally validated here. Speculative future applications include low-coverage atlas completion, simulation of treatment-response hypotheses, and condition-specific DE analysis in which generated cells are judged by downstream concordance rather than by distributional similarity alone; each would require dedicated experimental validation before claims of practical utility can be made.

Limitations. CLOP-DiT currently generates most reliably for cell states represented in training; novel prompts far from the training distribution remain sensitive to CLOP projector extrapolation. The 8 held-out validation studies share tissue types and cell populations with the 72 training studies, providing a study-level generalization test but not a stringent out-of-distribution evaluation. The cross-dataset biological validation (Table 2) compounds this: 4 of 5 datasets are from the training split. A pilot OOD/free-form prompt evaluation was executed (20 prompts spanning novel cell types, free-form descriptions, and cross-tissue contexts). Embedding-level analysis reveals that the CLOP conditioning vectors do differentiate prompts (pairwise cosine similarity 0.73–0.94 across semantically distinct prompts), and DiT-generated latent representations diverge further (cosine similarity 0.39–0.54). However, the frozen scGPT decoder collapses this latent variation: decoded expression profiles are near-identical across all tested prompts (expression cosine similarity ≈ 1.0). This confirms that the decoder—not the generation pipeline—dominates the expression output, and that gene-level fidelity metrics reflect decoder reconstruction rather than generative quality. Robust quantitative OOD scoring at the expression level is therefore not achievable with the current decoder architecture; the main reported expression metrics remain in-distribution and decoder-dominated. All generation and generalization claims should be interpreted accordingly. The absolute accuracy gap from real data and discriminator AUC above 0.5 suggest opportunities for improvement via larger DiT architectures, hard-negative mining in CLOP training, or multi-resolution conditioning schemes. All results derive from a single training run (seed 42); inter-run variance from stochastic initialization, data shuffling, and mini-batch ordering is uncharacterized. The bootstrap confidence intervals reported in Table S5 capture evaluation sampling variability only—they do not bound training-run variance. Reporting mean $\pm$ standard deviation over ≥ 3 independent seeds is a prerequisite for stronger statistical claims and is planned for a future revision.

Baseline scope. The benchmarking suite compares CLOP-DiT primarily against synthetic controls (Gaussian, shuffled, random, mean-collapse) and a single minimally-competitive learned baseline (EmbeddingVAE). An scVI baseline (scvi-tools v1.3.3, 200 epochs, 32-dimensional latent, seed 42) was trained on the same expression matrix and is included in the benchmark report. More broadly, the current baselines do not include a conditional VAE (e.g., scGen), a conditional GAN, or another diffusion-based model trained with label conditioning on the same data. The composite benchmark score (Section 2.5) assigns $s = 0$ to methods lacking downstream biology metrics; because CLOP-DiT is currently the only method evaluated on those axes, it receives the maximum normalized score on 8 of 17 metrics by construction, inflating its composite from 0.842 relative to what a common-metrics-only comparison would yield (0.702 for CLOP-DiT versus 0.750 for the Gaussian baseline, reversing the ranking). A "common-metrics-only" composite restricted to the 9 distributional metrics shared by all methods is reported alongside the full composite in the benchmark report to enable transparent like-for-like comparison. Future work should add at least one additional expression-space generative model (e.g., scGen, scArches [43], or CellOT [44]), and verify that CLOP-DiT's advantage persists under fair multi-pipeline comparison.

Marker gene and template caveats. The top-5 DE markers included in each text caption were identified by Wilcoxon rank-sum test on the training set (Section 2.1), creating a potential circularity: the model's text condition explicitly lists the genes that most distinguish each cell type in the training data, which may allow the model to reconstruct training-set DE patterns without learning generalizable regulatory programs. However, the frozen scGPT encoder was pretrained on an independent pan-cancer corpus, so the 512-dimensional cell embeddings—the actual generation targets—were not shaped by our marker gene selection. The markers appear only in the text conditioning signal and could in principle be exploited by the BiomedBERT encoder during CLOP training. A marker-dropped prompt ablation (removing all marker genes from the text template) and cross-referencing against an independent marker database (e.g., PanglaoDB [51] or CellMarker) would provide stronger evidence that the model is not merely memorizing training-set DE signatures. A pilot marker-dropped ablation was conducted on four cell types ($CD8^+$ T cells, macrophages, epithelial cells, NK cells): removing tissue context and all marker-gene mentions changes the CLOP conditioning vectors (cosine similarity to default prompt: 0.96–0.99) and DiT latent embeddings (cosine similarity: 0.91–0.97), confirming that the marker content does modulate the conditioning signal. However, the frozen scGPT decoder collapses these latent differences entirely (expression cosine similarity ≈ 1.0), preventing assessment of marker influence at the gene-expression level. A stronger test would require a trainable decoder or a decoder-free evaluation on the latent embeddings. Additionally, the text descriptions follow a rigid template rather than free-form natural language; the model may be mapping concatenated categorical tags to latent representations rather than learning language semantics. The novelty relative to label-conditioned generation lies in the richer conditioning signal (tissue, organism, markers, disease context beyond a bare cell-type label), but free-form language understanding is not demonstrated.

Loss function ablation. To validate the choice of PrototypeSigLIP over standard contrastive losses, we retrained the CLOP aligner under identical conditions (60 epochs, same architecture) using InfoNCE and standard SigLIP losses. InfoNCE [45] achieved only 3.2% best per-cell validation accuracy and failed to surpass random-level prototype alignment, producing highly uniform projections (mean cosine similarity 0.96) that preserve little inter-type variation. SigLIP improved moderately to 7.1% per-cell accuracy with better inter-group separation (1.00 vs. 0.91 for InfoNCE) but still lacked explicit text-to-group alignment. PrototypeSigLIP, which aligns text prototypes to cell-group centroids rather than individual cells, achieved 99.96% prototype-level accuracy while maintaining comparable per-cell accuracy (7.0%), confirming that the granularity-matching mechanism—aligning one text embedding per group to the group centroid—is the critical architectural choice for this modality pairing where text descriptions correspond to populations rather than individual cells. Separately, the component ablation (Table 3) reveals that removing cohesion regularization or cell noise augmentation raises CLOP per-cell validation accuracy by approximately 10 percentage points. This counterintuitive result reflects overfitting to the batch-level contrastive signal: without regularization the projector maps cells more tightly toward group centroids, inflating validation accuracy at the expense of representational diversity. The production configuration retains both regularizers because they produce better downstream generation quality and more diverse DiT outputs—a trade-off between contrastive proxy metrics and downstream task performance that is well recognized in representation learning.

The training corpus is restricted to 80 human and mouse datasets from cancer and developmental biology contexts. The pipeline has not been validated on other organisms (e.g., zebrafish), on healthy adult tissue atlases, or on perturbation experiments (e.g., Perturb-seq [56], drug treatments, CRISPR screens). Extending the corpus to multi-organism and multi-condition data would establish broader utility.

A pilot rare-cell augmentation experiment yielded negative results: generated cells provided marginal benefit for abundant cell types and no improvement for rare types (e.g., cycling cell F1 dropped from 0.83 to 0.79 with augmentation, and further to 0.74 at higher augmentation ratios). This outcome is consistent with the substantial fidelity gap from real data documented above, and indicates that augmentation-oriented applications will require further model improvements before practical deployment.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for text-conditioned single-cell gene expression generation. By combining contrastive language–omics pretraining (CLOP, 3.02M parameters) with a conditional diffusion transformer (DiT1D, 22.10M parameters) trained via flow matching, CLOP-DiT generates biologically meaningful cell profiles from structured text descriptions. Across 100 cell types from 80 datasets (220,304 cells), the model achieves 36.9% KNN accuracy (37$\times$ random chance, but 52 points below the 89% real-data ceiling) and 81% steering accuracy in the high-fidelity regime (CFG = 2.0, Euler), and a diversity ratio of 0.93 with the recommended Midpoint solver (CFG = 1.0). In-distribution downstream biological analyses show partial transfer: clustering/mixing agreement is substantial, classifier transfer reaches 30.8% accuracy (macro F1 0.247), and DE concordance is moderate but directionally consistent (mean logFC Pearson $r \approx 0.39$, mean sign agreement ≈ 0.94 across three tested contrasts). Decoder-reconstructed gene-mean fidelity is high ($r > 0.999$), but per-gene variance correlation is near zero ($r_{\text{var}} \approx 0$), indicating that cell-to-cell heterogeneity is not preserved—a critical limitation for single-cell applications that depend on within-population variability. CLOP-DiT therefore demonstrates a viable—but not yet indistinguishable from real data—text-conditioned generation paradigm for *in silico* single-cell biology (discriminator AUC = 0.656). All reported results are in-distribution and derive from a single training seed; multi-seed validation and stronger external baselines are needed before claims of general utility can be made.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales $\times$ 2 solvers $\times$ 3 metrics); Table S4: biological validation datasets; Table S5: bootstrap 95% confidence intervals (B=1,000, percentile method) for headline generation metrics. Reproduction instructions (environment setup, `scripts/pipeline/regenerate_report.sh`, expected outputs) are provided in `docs/operational/QUICK_START.md` and `scripts/pipeline/regenerate_report.sh`. All figures use colormaps designed for accessibility under common forms of color vision deficiency (deuteranopia, protanopia); critical distinctions additionally use shape or pattern encoding.

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Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop.yaml` and is deterministic given the dataset identifiers. Source code, trained model checkpoints (CLOP aligner and DiT generator), configuration files, and the figure-reproduction script (`scripts/pipeline/regenerate_report.sh`) are publicly available on GitHub [46] and will be archived on Zenodo upon acceptance of this manuscript. The complete software environment is specified in `requirements.txt` (Python 3.10, PyTorch 2.1, CUDA 12.0, scGPT v0.2.1, Transformers 4.36). One-command figure reproduction from cached embeddings is documented in `docs/operational/QUICK_START.md`.

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Abbreviations

The following abbreviations are used in this manuscript:

AdaLN	Adaptive Layer Normalization
ALL	Acute Lymphoblastic Leukemia
ALS	Amyotrophic Lateral Sclerosis
AML	Acute Myeloid Leukemia
AMP	Automatic Mixed Precision
ARI	Adjusted Rand Index
ASD	Autism Spectrum Disorder
AUC	Area Under Curve
BCC	Basal Cell Carcinoma
BERT	Bidirectional Encoder Representations from Transformers
CE	Cross-Entropy
CFG	Classifier-free guidance
CI	Confidence Interval
CLIP	Contrastive Language-Image Pre-training
CLOP	Contrastive Language-Omics Pretraining
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CUDA	Compute Unified Device Architecture
DE	Differential expression
DiT	Diffusion Transformer
DivR	Diversity Ratio
EMA	Exponential Moving Average
ESC	Embryonic Stem Cell
FD	Fréchet Distance
FP16	Half-Precision Floating Point
GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GEO	Gene Expression Omnibus
GPU	Graphics Processing Unit
HBV	Hepatitis B Virus
IFN	Interferon
InfoNCE	Info Noise-Contrastive Estimation
KNN	*k*-Nearest Neighbor
LinAcc	Linear Classifier Accuracy
logFC	Log Fold Change
LoRA	Low-Rank Adaptation
LPS	Lipopolysaccharide
LR	Learning Rate
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
NK	Natural Killer
NMI	Normalized Mutual Information
ODE	Ordinary Differential Equation
OOD	Out-of-Distribution
PBMC	Peripheral Blood Mononuclear Cell
PCA	Principal Component Analysis
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
scGPT	Single-cell Generative Pre-trained Transformer
scRNA-seq	Single-cell RNA sequencing
SigLIP	Sigmoid Loss for Language-Image Pre-training
UMAP	Uniform Manifold Approximation and Projection
VAE	Variational Autoencoder
VRAM	Video Random Access Memory
ZCA	Zero-phase Component Analysis

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→512]	1.58M
CLOP cell projector	MLP [512→512→512→512]	1.44M
DiT1D	8× AdaLN-Zero blocks, 8 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 100 evaluation groups. Note that these values are computed on a different subsample (100 groups × 200 cells) than the primary operating-point results in Table 1 (100 types × 200 cells with different random seeds), which accounts for minor numerical differences between the two tables.

Table A2. CFG scale sweep (10-step Euler, 100 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

Appendix C. Training Hyperparameters

Table A3 lists the complete training hyperparameters for both CLOP alignment and DiT flow-matching stages, corresponding to `configs/clop.yaml` and `configs/dit.yaml`.

Appendix D. Supplementary Tables

Table A4. All 80 GEO datasets used in CLOP-DiT training and validation. Datasets were preprocessed to 2,000 highly variable genes per dataset. Split: Train or held-out Validation (study-level, seed 42).

#	Accession	Description	n_{cells}	Split
1	GSE103867	scRNA-seq (H. sapiens)	2,787	Val
2	GSE110686	scRNA-seq (H. sapiens)	2,992	Train
3	GSE115571	Mouse cells, LPS stimulation	442	Train
4	GSE117988	PBMCs from patients	2,638	Train
5	GSE117988	Merkel cell carcinoma tumor	2,990	Train
6	GSE120505	Aged blood cells	3,000	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
7	GSE123813	Human basal cell carcinoma	3,000	Train
8	GSE123813	Human cutaneous squamous cell carcinoma	3,000	Train
9	GSE123902	Human lung adenocarcinoma	2,753	Train
10	GSE124310	Human multiple myeloma bone marrow	2,833	Train
11	GSE130148	Human fetal lung development	3,000	Train
12	GSE132509	Pediatric bone marrow mononuclear cells	2,984	Train
13	GSE138709	Human intrahepatic cholangiocarcinoma	2,891	Train
14	GSE142653	Human pituitary gland development	2,885	Train
15	GSE143423	Human leptomeningeal brain metastasis	3,000	Train
16	GSE143423	Triple-neg. breast cancer brain met.	2,332	Train
17	GSE145929	Adult mouse prostate	2,830	Train
18	GSE145929	Adult mouse urethra	2,745	Train
19	GSE148218	Human bone marrow, ALL	2,807	Train
20	GSE149655	Human early-stage lung adenocarcinoma	2,087	Train
21	GSE155109	Endothelial cells, breast cancer	3,000	Train
22	GSE155109	Stromal cells, breast cancer	3,000	Train
23	GSE163558	Human stomach cancer	2,467	Train
24	GSE165784	Human retina development	2,622	Train
25	GSE165844	Mouse LSK hematopoietic progenitors	2,881	Train
26	GSE167597	Mouse spinal cord development	3,000	Val
27	GSE168181	Human breast cancer tissue	2,863	Train
28	GSE175975	scRNA-seq (H. sapiens)	2,728	Train
29	GSE183904	Human gastric cancer	3,000	Val
30	GSE189070	Mouse astrocytes, spinal cord injury	2,960	Train
31	GSE189357	Human lung adenocarcinoma	2,886	Train
32	GSE192857	hESC differentiation time-series	2,696	Train
33	GSE212502	Human forebrain organoids	3,000	Train
34	GSE213740	Human ascending aortic wall	2,704	Val
35	GSE220913	Human HL-60 cell lines, CFTR	2,972	Train
36	GSE222002	T cells from human cancer	2,962	Train
37	GSE222369	NK cells, human lymphoma	2,778	Train
38	GSE225600	Human breast cancer tumor	2,333	Train
39	GSE225857	Colorectal cancer liver metastases	2,986	Train
40	GSE226131	Aged mouse hematopoietic stem cells	2,608	Train
41	GSE226762	scRNA-seq (H. sapiens)	830	Val
42	GSE227644	scRNA-seq (H. sapiens)	3,000	Train
43	GSE227719	Mouse KrasG12D/P53 lung organoids	2,990	Train
44	GSE228499	Human breast cancer	2,448	Train
45	GSE235787	B cells, ALL	2,798	Train
46	GSE236565	Mouse splenic CD4+ T cells	2,556	Train
47	GSE248214	scRNA-seq (H. sapiens)	2,998	Train
48	GSE253355	Human bone marrow niche	2,835	Train
49	GSE262288	Metastatic breast cancer	2,321	Train
50	GSE272187	scRNA-seq (H. sapiens)	2,639	Train
51	GSE275119	Mouse dental tissue development	2,963	Train
52	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644	Val
53	GSE279781	scRNA-seq (H. sapiens)	2,654	Train
54	GSE280847	CD45+ leukocytes, mouse MC38 tumor	2,864	Train
55	GSE283205	Hepatoblastoma tumor tissue	2,994	Train
56	GSE283397	Human CD8+ T cells, chronic HBV	2,856	Train
57	GSE288211	Mouse pancreatic islets, Zzef1 KO	2,724	Train
58	GSE289611	Human pulmonary pleomorphic carcinoma	2,983	Train
59	GSE291166	Mouse mesenteric lymph node	2,901	Train
60	GSE299623	scRNA-seq (H. sapiens)	2,991	Train
61	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716	Val
62	GSE302433	Mouse B16F10 melanoma, ZDHHC13	2,953	Val
63	GSE303309	Brain CD45+ immune cells, Alzheimer's	2,999	Train
64	GSE306567	Mouse ESCs and neural progenitors	2,454	Train
65	GSE306676	Mouse spinal cord, SOD1-G93A ALS	2,959	Train
66	GSE307261	Human B and T cells, melanoma	2,921	Train
67	GSE307774	Mouse colorectal tumors, Kras/Braf	2,505	Train
68	GSE308428	Mouse hippocampal microglia, ASD	2,991	Train
69	GSE309368	Human myeloid cells, healthy donors	1,361	Train
70	GSE315628	scRNA-seq (M. musculus)	2,895	Train
71	GSE318353	scRNA-seq (H. sapiens)	2,964	Train
72	GSE98638	Human hepatocellular carcinoma T cells	3,000	Train
73	GSE120446	Bone marrow, hematopoietic development	2,890	Train
74	GSE104323	Dentate gyrus, brain development	3,000	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
75	GSE75748	Endoderm differentiation	2,531	Train
76	GSE144024	Human embryonic stem cells	3,000	Train
77	GSE72857	Hematopoiesis	3,000	Train
78	GSE226824	IFN-treated hematopoietic progenitors	2,739	Train
79	GSE141259	Lung development	3,000	Train
80	GSE139369	Human hematopoiesis (Setty et al.)	2,995	Train
	Total		**220,304**	

Table A6. Full CFG scale sweep. 200 cells generated per group (100 groups). KNN: type-identity accuracy; Steer.: alignment; DivR: diversity ratio (1.0 = ideal); LinAcc: linear classifier accuracy; FD: Fréchet distance.

CFG	Solver	Steps	KNN-1	KNN-5	Steer.	DivR	LinAcc	FD
Primary configurations (Table 1)								
3.0	Euler	10	0.367	0.554	0.802	0.473	0.508	3.46
2.0	Euler	10	0.369	0.553	0.810	0.513	0.511	3.17
1.0	Euler	10	0.313	0.483	0.810	0.745	0.390	2.84
0.0	Euler	10	0.010	0.051	0.475	1.833	0.010	0.58
3.0	Midpoint	10	0.340	0.528	0.792	0.628	0.477	3.38
2.0	Midpoint	10	0.340	0.525	0.800	0.686	0.475	3.13
1.0	Midpoint	10	0.288	0.461	0.807	0.929	0.357	2.64
3.0	Euler	50	0.348	0.545	0.795	0.622	0.489	3.33
1.0	Euler	50	0.293	0.466	0.807	0.919	0.365	2.65
—	Gaussian	—	0.011	0.048	0.466	2.277	0.009	0.03
Fine-grained CFG sweep (Euler solver)								
0.5	Euler	10	0.147	0.284	0.788	1.129	—	2.33
1.0	Euler	10	0.348	0.515	0.796	0.757	—	2.94
1.5	Euler	10	0.396	0.553	0.798	0.600	—	3.36
2.0	Euler	10	0.407	0.580	0.806	0.526	—	3.54
2.5	Euler	10	0.408	0.590	0.804	0.493	—	3.70
3.0	Euler	10	0.410	0.592	0.802	0.484	—	3.80
4.0	Euler	10	0.402	0.586	0.804	0.502	—	4.01
5.0	Euler	10	0.380	0.576	0.798	0.548	—	4.34
0.5	Euler	20	0.137	0.269	0.788	1.230	—	2.07
1.0	Euler	20	0.334	0.504	0.794	0.842	—	2.83
1.5	Euler	20	0.380	0.545	0.798	0.679	—	3.19
2.0	Euler	20	0.391	0.569	0.798	0.600	—	3.43
2.5	Euler	20	0.394	0.578	0.800	0.559	—	3.66
3.0	Euler	20	0.398	0.582	0.800	0.540	—	3.77
4.0	Euler	20	0.388	0.578	0.802	0.535	—	3.97
5.0	Euler	20	0.379	0.567	0.792	0.551	—	4.16
0.5	Euler	50	0.132	0.259	0.792	1.318	—	1.95
1.0	Euler	50	0.324	0.495	0.794	0.933	—	2.72
1.5	Euler	50	0.370	0.538	0.796	0.776	—	3.14
2.0	Euler	50	0.384	0.563	0.796	0.699	—	3.31
2.5	Euler	50	0.388	0.576	0.796	0.658	—	3.48
3.0	Euler	50	0.392	0.581	0.792	0.637	—	3.64
4.0	Euler	50	0.379	0.579	0.790	0.622	—	3.86
5.0	Euler	50	0.378	0.570	0.790	0.626	—	4.10

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Table A3. Training hyperparameters.

Hyperparameter	CLOP	DiT
Optimizer	AdamW	AdamW (β_1=0.9, β_2=0.999)
Learning rate	5×10^{-4}	2×10^{-4}
Weight decay	0.06	0.01
Batch size	1024	1024
Epochs	60	200
Warmup epochs	5	5
LR schedule	Cosine with warmup	Cosine with warmup
Gradient clipping	1.0	1.0
Mixed precision	FP16 (AMP)	FP16 (AMP)
Dropout	0.2	Proj: 0.1, Attn: 0.0
EMA decay	—	0.9999
Temperature	14.0 (fixed)	—
Loss	PrototypeSigLIP	Flow matching MSE
Cohesion weight	0.15	—
Label smoothing	0.1	—
CFG drop prob	—	0.15
Time sampling	—	Logit-normal (μ=0, σ=1)

Table A5. The 8 held-out validation datasets (study-level split, seed 42, val_split = 0.1). No cells from these studies were seen during CLOP or DiT training.

#	Accession	Description	n_{cells}
1	GSE103867	scRNA-seq (H. sapiens)	2,787
2	GSE167597	Mouse spinal cord development	3,000
3	GSE183904	Human gastric cancer	3,000
4	GSE213740	Human ascending aortic wall	2,704
5	GSE226762	scRNA-seq (H. sapiens)	830
6	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644
7	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716
8	GSE302433	Mouse B16F10 melanoma, ZD-HHC13	2,953
	Total		**20,634**

Table A7. The 5 biological validation datasets used for marker gene correlation, expression distribution fidelity, and Gene Ontology enrichment analysis (Section 3.10). These datasets were selected to span diverse human cancer tissue types and immune cell populations. Four of the five datasets are from the training split; GSE183904 is from the held-out validation set (Table S2).

#	GEO Accession	Tissue	Description	Split	n_{ce}
1	GSE123902	Lung	Lung adenocarcinoma tumor microenvironment	Train	2,7
2	GSE183904	Gastric	Gastric cancer tumor microenvironment	Val	3,00
3	GSE123813	Skin	Basal cell carcinoma of the skin	Train	3,00
4	GSE98638	Liver	T cell landscape of hepatocellular carcinoma	Train	3,00
5	GSE132509	Blood	Acute lymphoblastic leukemia bone marrow cells	Train	2,9
	Total				**14,7**

Table A8. Bootstrap 95% confidence intervals (percentile method, B=1,000) for headline generation metrics. Resampling unit: cell types (n=69 types resampled with replacement). Classifiers (KNN, logistic regression) were trained once on real data; only the composition of evaluated types varies across bootstrap iterations. Generation configuration: CFG = 1.5, Midpoint solver, 20 steps. Point estimates reflect the full (non-resampled) evaluation over all 69 types (100 generated cells per type, 6,900 total).

Metric	Point Est.	95% CI Lower	95% CI Upper	Boot. SE
KNN Top-1 Accuracy	0.509	0.421	0.586	0.043
KNN Top-5 Accuracy	0.728	0.658	0.794	0.035
Steering Accuracy	1.000	1.000	1.000	0.000
Diversity Ratio (DivR)	0.969	0.963	0.974	0.003
Linear Classifier Acc.	0.583	0.518	0.646	0.034
Centroid Cosine Sim.	0.898	0.875	0.913	0.009

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